第十七章 隐函数存在定理

§1 单个方程的情形

1．设函数$F(x,y)$满足

(1) 在区域$D:x_0-a\le x\le x_0+a,\ y_0-b\le x\le y_0+b$上连续；

(2) $F(x_0,y_0)=0$；

(3) 当x固定时，函数$F(x,y)$是y的严格单调函数；

则可得到什么结论？试证明之.

解 由已知条件，可得结论

(i) 存在函数$y=f(x)$定义在$(x_0-\alpha,x_0+\alpha)$内，满足$F(x,f(x))\equiv 0$，且$y_0=f(x_0)$；

(ii) 函数$y=f(x)$在$(x_0-\alpha,x_0+\alpha)$内连续.

下面进行证明

(i) 由条件(3),当x固定时，函数$F(x,y)$是y的严格单调函数，不妨设$F(x,y)$关于y严格单调递.

固定$x=x_0$，由条件(2)知$F(x_0,y_0)=0$，从而

$$F(x_0,y_0-b)<0,\ F(x_0,y_0+b)>0$$

在$F(x,y)$中分别固定$y=y_0-b$和$y=y_0+b$，由一元函数$F(x,y_0-b)$和$F(x,y_0+b)$在x_0连续，以及连续函数的保号性，知$\exists\alpha_1>0\,(\alpha_1<a)$，使得当$x\in(x_0-\alpha_1,x_0+\alpha_1)$时有

$$F(x,y_0-b)<0 \tag{1}$$

$\exists\alpha_2>0\,(\alpha_2<a)$，使得当$x\in(x_0-\alpha_2,x_0+\alpha_2)$时有

$$F(x,y_0+b)>0 \tag{2}$$

取$\alpha=\min(\alpha_1,\alpha_2)$，则当$x\in(x_0-\alpha,x_0+\alpha)$时，（1）（2）两式同时成立，对任意$\bar{x}\in(x_0-\alpha,x_0+\alpha)$，一元函数$F(\bar{x},y)$在$y\in(y_0-b,y_0+b)$连续，并且

$$F(\bar{x},y_0-b)<0,\ F(\bar{x},y_0+b)>0$$

根据一元函数的介值定理，$\exists \bar{y} \in (y_0 - b, y_0 + b)$，使得 $F(\bar{x}, \bar{y}) = 0$.

又因为 $F(\bar{x}, y)$ 关于 y 在 $[y_0 - b, y_0 + b]$ 严格单调上升，故上述 $\bar{y}$ 是唯一的，这样就确定了一个定义在区间 $(x_0 - \alpha, x_0 + \alpha)$ 上的隐函数 $y = f(x)$，特别地 $y_0 = f(x_0)$，这样就证明了结论(i)；

（ii）任给 $\bar{x} \in (x_0 - \alpha, x_0 + \alpha)$，记 $\bar{y} = f(\bar{x})$，下证 $f(x)$ 在 $\bar{x}$ 连续.

对 $\forall \varepsilon > 0$，不妨让 ε 充分小使得 $[\bar{y} - \varepsilon, \bar{y} + \varepsilon] \subset [y_0 - b, y_0 + b]$，因为 y 的一元函数 $F(\bar{x}, y)$ 在 $[\bar{y} - \varepsilon, \bar{y} + \varepsilon]$ 上严格单调上升且 $F(\bar{x}, \bar{y}) = 0$，所以

$$F(\bar{x}, \bar{y} - \varepsilon) < 0, F(\bar{x}, \bar{y} + \varepsilon) > 0$$

而 x 的一元函数 $F(x, \bar{y} - \varepsilon)$ 和 $F(x, \bar{y} + \varepsilon)$ 在 $x \in (x_0 - \alpha, x_0 + \alpha)$ 连续，故 $\exists \delta_1 > 0$，满足 $(\bar{x} - \delta_1, \bar{x} + \delta_1) \subset (x_0 - \alpha, x_0 + \alpha)$，而且当 $x \in (\bar{x} - \delta_1, \bar{x} + \delta_1)$ 时，有

$$F(x, \bar{y} - \varepsilon) < 0 \tag{3}$$

$\exists \delta_2 > 0$，满足 $(\bar{x} - \delta_2, \bar{x} + \delta_2) \subset (x_0 - \alpha, x_0 + \alpha)$，而且当 $x \in (\bar{x} - \delta_2, \bar{x} + \delta_2)$ 时，有

$$F(x, \bar{y} + \varepsilon) > 0 \tag{4}$$

取 $\delta = \min\{\delta_1, \delta_2\} > 0$，则当 $x \in (\bar{x} - \delta, \bar{x} + \delta)$ 时，(3)(4)两式同时成立. 因此只要 $x \in (\bar{x} - \delta, \bar{x} + \delta)$，$F(x, y)$ 作为 y 的函数在 $(\bar{y} - \varepsilon, \bar{y} + \varepsilon)$ 就严格单调上升，且有唯一的零点 $y = f(x)$，显然满足 $y \in (\bar{y} - \varepsilon, \bar{y} + \varepsilon)$，即 $|f(x) - f(\bar{x})| < \varepsilon$，从而结论（ii）得证.

2．方程 $x^2 + y + \sin(xy) = 0$ 在原点附近能否用形如 $y = f(x)$ 的隐函数表示？又能否用形如 $x = g(y)$ 的隐函数表示？

解 令 $F(x) = x^2 + y + \sin(xy)$，则 $F(0,0) = 0$ 并且

$$F_x = 2x + y\cos(xy),\ F_y = 1 + x\cos(xy)$$

它们都在全平面上连续，而且 $F_y(0,0) = 1$，因而方程在 $(0,0)$ 点的邻域内可唯一地确定可微函数的隐函数 $y = f(x)$，但由于 $F_x(0,0) = 0$，因而据此无法判定是否在 $(0,0)$ 点的某

邻域内有隐函数 $x=g(y)$ 存在.

3. 方程 $F(x,y)=y^2-x^2(1-x^2)=0$ 在哪些点的附近可以唯一地确定单值、连续且有连续导数的函数 $y=f(x)$.

解 $F_x(x,y)=4x^3-2x, F_y(x,y)=2y$ 均在全平面连续，而且 $F_y(x,y)\big|_{y\neq 0}\neq 0$，因而在方程 $F(x,y)=0$ 的除去 $(0,0)$，$(\pm 1,0)$ 的解点处，均可唯一地确定单值、连续、且有连续导数的函数 $y=f(x)$.

4. 证明有唯一可导函数 $y=y(x)$ 满足方程 $\sin y+\sinh y=x$，并求出导数 $y'(x)$，其中 $\sinh y=\dfrac{e^y-e^{-y}}{2}$.

证明 设 $F(x,y)=\sin y+\sinh y-x$，则 $F_x(x,y)=-1$ 和 $F_y(x,y)=\cos y+\cosh y$ 在全平面连续. 显然当 $y=0$ 时 $F_y(x,y)=\cos y+\cosh y>0$；当 $y\neq 0$ 时，根据平均值不等式 $\cosh y=\dfrac{e^y+e^{-y}}{2}\geq\sqrt{e^ye^{-y}}=1$，也可以得到 $F_y(x,y)=\cos y+\cosh y>0$. 因而在方程 $F(x,y)=\sin y+\sinh y-x=0$ 的任一解点附近可确定唯一的可导的函数 $y=y(x)$，且

$$y'(x)=-\frac{F_x(x,y)}{F_y(x,y)}=\frac{1}{\cos y+\cosh y}$$

5. 方程 $xy+z\ln y+e^{xz}=1$ 在点 $P_0(0,1,1)$ 的某邻域内能否确定出某一个变量是另外两个变量的函数.

解 设 $F(x,y,z)=xy+z\ln y+e^{xz}-1$，则 $F(0,1,1)=0$，而且

$$F_x(x,y,z)=y+ze^{xz}, F_y(x,y,z)=x+\frac{z}{y}, F_z(x,y,z)=\ln y+xe^{xz}$$

均在全平面连续，又 $F_x(0,1,1)=2\neq 0$，$F_y(0,1,1)=1\neq 0$，$F_z(0,1,1)=0$，因此在点 $P_0(0,1,1)$ 的某邻域内，可以确定出 $x=x(y,z)$，亦可确定出 $y=(x,z)$，但由于

$F_z(0,1,1)=0$，据此无法确定是否在 $P_0(0,1,1)$ 点的某邻域内有隐函数 $z=z(x,y)$ 存在.

6. 设 f 是一元函数，试问 f 应满足什么条件，方程 $2f(xy)=f(x)+f(y)$ 在点 $(1,1)$ 的邻域内能否确定出唯一的 y 是 x 的函数.

解 设 $F(x,y)=2f(xy)-f(x)-f(y)$，则 $F(1,1)=0$，且当 f 连续可导时，有

$$F_x(x,y)=2yf'(xy)-f'(x),\quad F_y(x,y)=2xf'(xy)-f'(y)$$

它们在 $(1,1)$ 邻域内连续，而且 $F_x(1,1)=2f'(1)-f'(1)=f'(1)$，$F_y(1,1)=f'(1)$，因而只要 $f'(1)\neq 0$ 时，就有 $F_y(1,1)\neq 0$，这时方程 $2f(xy)=f(x)+f(y)$ 在点 $(1,1)$ 的邻域内能确定出唯一的 y 为 x 的函数.

通过上述分析知，当 f 在 $x_0=1$ 的某邻域内有连续的一阶导数且 $f'(1)\neq 0$ 时，方程 $2f(x,y)=f(x)+f(y)$ 在点 $(1,1)$ 的邻域内能确定出唯一的 y 为 x 的函数.

7. 设有方程 $x=y+\varphi(y)$，其中 $\varphi(0)=0$，且当 $-a<y<a$ 时，$|\varphi'(y)|\leq k<1$. 证明：存在 $\delta>0$，当 $-\delta<x<\delta$ 时，存在唯一的可微函数 $y=y(x)$ 满足方程 $x=y+\varphi(y)$ 且 $y(0)=0$.

证明 设 $F(x,y)=x-y-\varphi(y)$，则 $F(0,0)=0$，且

$$F_x(x,y)=1,\quad F_y(x,y)=-1-\varphi'(y)\neq 0$$

由 $F_y(x,y)=-1-\varphi'(y)\neq 0$ 知函数 $F(x,y)$ 关于 y 在 $(0,0)$ 的某领域内严格单调，因而由本节习题 1 知 $\exists\delta>0$，当 $-\delta<x<\delta$ 时，存在唯一的可微函数 $y=y(x)$，满足方程 $x=y+\varphi(y)$ 且 $y(0)=0$.

§2 方程组的情形

1. 试讨论方程组

$$\begin{cases}x^2+y^2=\frac{1}{2}z^2\\ x+y+z=2\end{cases}$$

在点 $P_0(1,-1,2)$ 的附近能否确定形如 $x=f(z)$，$y=g(z)$ 的隐函数组.

解 令

$$\begin{cases} F(x,y,z)=x^2+y^2-\frac{1}{2}z^2 \\ G(x,y,z)=x+y+z-2 \end{cases}$$

显然 $F(x,y,z)$ 和 $G(x,y,z)$ 在全平面有连续的偏导数，故它们在点 $P_0(1,-1,2)$ 的附近固然也有连续偏导数，而且 $F(1,-1,2)=0$，$G(1,-1,2)=0$，又因为

$$\left.\frac{\partial(F,G)}{\partial(x,y)}\right|_{P_0}=\left.\begin{vmatrix} F_x & F_y \\ G_x & G_y \end{vmatrix}\right|_{P_0}=\left.\begin{vmatrix} 2x & 2y \\ 1 & 1 \end{vmatrix}\right|_{P_0}=\begin{vmatrix} 2 & -2 \\ 1 & 1 \end{vmatrix}=4\neq 0$$

因此在点 $P_0(1,-1,2)$ 的某邻域内方程组可唯一地确定形如 $x=f(z)$，$y=g(z)$ 隐函数组.

2. 求下列函数组的反函数组的偏导数：

(1) 设 $u=x\cos\frac{y}{x},v=x\sin\frac{y}{x}$，求 $\frac{\partial x}{\partial u},\frac{\partial x}{\partial v},\frac{\partial y}{\partial u},\frac{\partial y}{\partial v}$；

(2) 设 $u=e^x+x\sin y,v=e^x-x\cos y$，求 $\frac{\partial x}{\partial u},\frac{\partial x}{\partial v},\frac{\partial y}{\partial u},\frac{\partial y}{\partial v}$.

解 (1) 由于 $u=x\cos\frac{y}{x},v=x\sin\frac{y}{x}$，故

$$\frac{\partial u}{\partial x}=\cos\frac{y}{x}+\frac{y}{x}\sin\frac{y}{x},\quad \frac{\partial u}{\partial y}=-\sin\frac{y}{x},\quad \frac{\partial v}{\partial x}=\sin\frac{y}{x}-\frac{y}{x}\cos\frac{y}{x},\quad \frac{\partial v}{\partial y}=\cos\frac{y}{x}$$

即函数组 $u=x\cos\frac{y}{x},v=x\sin\frac{y}{x}$ 在 $x\neq 0$ 处对 x,y 的偏导数是连续的，又由于

$$J=\frac{\partial(u,v)}{\partial(x,y)}=\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}=\begin{vmatrix} \cos\frac{y}{x}+\frac{y}{x}\sin\frac{y}{x} & -\sin\frac{y}{x} \\ \sin\frac{y}{x}-\frac{y}{x}\cos\frac{y}{x} & \cos\frac{y}{x} \end{vmatrix}=1\neq 0$$

因而由反函数组定理得

$$\frac{\partial x}{\partial u}=\frac{1}{J}\frac{\partial v}{\partial y}=\cos\frac{y}{x},\qquad \frac{\partial x}{\partial v}=-\frac{1}{J}\frac{\partial u}{\partial y}=\sin\frac{y}{x},$$

$$\frac{\partial y}{\partial u}=-\frac{1}{J}\frac{\partial v}{\partial x}=\frac{y}{x}\cos\frac{y}{x}-\sin\frac{y}{x},\quad \frac{\partial y}{\partial v}=\frac{1}{J}\frac{\partial u}{\partial x}=\frac{y}{x}\sin\frac{y}{x}+\cos\frac{y}{x}.$$

(2) 由 $u=e^x+x\sin y,v=e^x-x\cos y$ 的表达式知它们在全平面存在对 x,y 的连续偏导数，且

$$\frac{\partial u}{\partial x}=e^x+\sin y,\quad \frac{\partial u}{\partial y}=x\cos y,\quad \frac{\partial v}{\partial x}=e^x-\cos y,\quad \frac{\partial v}{\partial y}=x\sin y$$

又由于

$$J=\frac{\partial(u,v)}{\partial(x,y)}=\begin{vmatrix}\dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial y}\\ \dfrac{\partial v}{\partial x} & \dfrac{\partial v}{\partial y}\end{vmatrix}=\begin{vmatrix}e^x+\sin y & x\cos y\\ e^x-\cos y & x\sin y\end{vmatrix}=x\left(e^x(\sin y-\cos y)+1\right)$$

故在$J\neq 0$的任何点的邻域内，都有

$$\frac{\partial x}{\partial u}=\frac{1}{J}\frac{\partial v}{\partial y}=\frac{\sin y}{e^x\left(\sin y-\cos y\right)+1},\qquad \frac{\partial x}{\partial v}=-\frac{1}{J}\frac{\partial u}{\partial y}=-\frac{\cos y}{e^x\left(\sin y-\cos y\right)+1},$$

$$\frac{\partial y}{\partial u}=-\frac{1}{J}\frac{\partial v}{\partial x}=\frac{\cos y-e^x}{x\left[e^x(\sin y-\cos y)+1\right]},\quad \frac{\partial y}{\partial v}=\frac{1}{J}\frac{\partial u}{\partial x}=\frac{e^x+\sin y}{x\left[e^x(\sin y-\cos y)+1\right]}.$$

3. 设$u=\dfrac{x}{r^2},v=\dfrac{y}{r^2},w=\dfrac{z}{r^2}$，其中$r=\sqrt{x^2+y^2+z^2}$.

(1) 试求以u,v,w为自变量的反函数组；

(2) 计算$\dfrac{\partial(u,v,w)}{\partial(x,y,z)}$.

解 (1)根据已知条件$u=\dfrac{x}{r^2},v=\dfrac{y}{r^2},w=\dfrac{z}{r^2}$可得$x=ur^2,y=vr^2,z=wr^2$，将其代入公式$r=\sqrt{x^2+y^2+z^2}$知$r^2=u^2r^4+v^2r^4+w^2r^4$，化简得

$$r^2=\frac{1}{u^2+v^2+w^2}$$

因而

$$\begin{cases}x=ur^2=\dfrac{u}{u^2+v^2+w^2}\\ y=vr^2=\dfrac{v}{u^2+v^2+w^2}\\ z=wr^2=\dfrac{w}{u^2+v^2+w^2}\end{cases}$$

(2)根据u,v,w的表达式可得它们对x,y,z的各个偏导数，从而有

$$\frac{\partial(u,v,w)}{\partial(u,v,w)}=\begin{vmatrix}\frac{1}{r^2}-\frac{2x^2}{r^4} & -\frac{2xy}{r^4} & -\frac{2xz}{r^4}\\ -\frac{2xy}{r^4} & \frac{1}{r^2}-\frac{2y^2}{r^4} & -\frac{2yz}{r^4}\\ -\frac{2xz}{r^4} & -\frac{2yz}{r^4} & \frac{1}{r^2}-\frac{2z^2}{r^4}\end{vmatrix}$$

$$=-\frac{1}{r^{12}}\begin{vmatrix}2x^2-r^2 & 2xy & 2xz\\ 2xy & 2y^2-r^2 & 2yz\\ 2xz & 2yz & 2z^2-r^2\end{vmatrix}=-\frac{1}{r^6}$$

4. 设 f_i， φ_i 连续可微，且

$$F_i(x_1,x_2,\cdots,x_n)=f_i(\varphi_1(x_1),\varphi_2(x_2),\cdots,\varphi_n(x_n))\quad (i=1,2,\cdots,n)$$

求 $\dfrac{\partial(F_1,F_2,\cdots,F_n)}{\partial(x_1,x_2,\cdots,x_n)}$.

解 将 $\varphi_1(x_1),\varphi_2(x_2),\cdots,\varphi_n(x_n)$ 看作中间变量，根据复合函数求导法则有

$$\frac{\partial(F_1,F_2,\cdots,F_n)}{\partial(x_1,x_2,\cdots,x_n)}=\begin{vmatrix}\frac{\partial F_1}{\partial x_1} & \frac{\partial F_1}{\partial x_2} & \cdots & \frac{\partial F_1}{\partial x_n}\\ \frac{\partial F_2}{\partial x_1} & \frac{\partial F_2}{\partial x_2} & \cdots & \frac{\partial F_2}{\partial x_n}\\ \cdots & \cdots & \cdots & \cdots\\ \frac{\partial F_n}{\partial x_1} & \frac{\partial F_n}{\partial x_2} & \cdots & \frac{\partial F_n}{\partial x_n}\end{vmatrix}=\begin{vmatrix}\frac{\partial f_1}{\partial\varphi_1}\frac{d\varphi_1}{dx_1} & \frac{\partial f_1}{\partial\varphi_2}\frac{d\varphi_2}{dx_2} & \cdots & \frac{\partial f_1}{\partial\varphi_n}\frac{d\varphi_n}{dx_n}\\ \frac{\partial f_2}{\partial\varphi_1}\frac{d\varphi_1}{dx_1} & \frac{\partial f_2}{\partial\varphi_2}\frac{d\varphi_2}{dx_2} & \cdots & \frac{\partial f_2}{\partial\varphi_n}\frac{d\varphi_n}{dx_n}\\ \cdots & \cdots & \cdots & \cdots\\ \frac{\partial f_n}{\partial\varphi_1}\frac{d\varphi_1}{dx_1} & \frac{\partial f_n}{\partial\varphi_2}\frac{d\varphi_2}{dx_2} & \cdots & \frac{\partial f_n}{\partial\varphi_n}\frac{d\varphi_n}{dx_n}\end{vmatrix}$$

$$=\frac{d\varphi_1}{dx_1}\cdot\frac{d\varphi_2}{dx_2}\cdot\cdots\cdot\frac{d\varphi_n}{dx_n}\begin{vmatrix}\frac{\partial f_1}{\partial\varphi_1} & \frac{\partial f_1}{\partial\varphi_2} & \cdots & \frac{\partial f_1}{\partial\varphi_n}\\ \frac{\partial f_2}{\partial\varphi_1} & \frac{\partial f_2}{\partial\varphi_2} & \cdots & \frac{\partial f_2}{\partial\varphi_n}\\ \cdots & \cdots & \cdots & \cdots\\ \frac{\partial f_n}{\partial\varphi_1} & \frac{\partial f_n}{\partial\varphi_2} & \cdots & \frac{\partial f_n}{\partial\varphi_n}\end{vmatrix}$$

$$=\varphi_1'(x_1)\varphi_2'(x_2)\cdots\varphi_n'(x_n)\frac{\partial(f_1,f_2,\cdots,f_n)}{\partial(\varphi_1,\varphi_2,\cdots,\varphi_n)}$$

5. 据理说明：在点 $(0,1)$ 附近是否存在连续可微函数 $f(x,y)$ 和 $g(x,y)$ 满足 $f(0,1)=1$， $g(0,1)=-1$，且

$$\left[f\left(x,y\right)\right]^3+xg\left(x,y\right)-y=0,$$

$$\left[g\left(x,y\right)\right]^3+yf\left(x,y\right)-x=0.$$

分析 令

$$\begin{cases}F(x,y,u,v)=u^3+xv-y\\G(x,y,u,v)=v^3+yu-x\end{cases}$$

则F,G关于各个变元在$P_0(0,1,1,-1)$附近有连续偏导数，又

$$F\left(0,1,1,-1\right)=0,\ \ G\left(0,1,1,-1\right)=0,$$

且$\left.\dfrac{\partial\left(F,G\right)}{\partial\left(u,v\right)}\right|_{(0,1,1,-1)}=\left.\begin{vmatrix}3u^2 & x\\ y & 3v^2\end{vmatrix}\right|_{(0,1,1,-1)}=9\neq0$，因而由隐函数存在定理，在点$\left(0,1\right)$附近存在连续可微函数$u=f\left(x,y\right)$和$v=g\left(x,y\right)$满足$f\left(0,1\right)=1$，$g\left(0,1\right)=-1$，且

$$\left[f\left(x,y\right)\right]^3+xg\left(x,y\right)-y=0,$$
$$\left[g\left(x,y\right)\right]^3+yf\left(x,y\right)-x=0.$$

6. 设

$$\begin{cases}u=f(x,y,z,t),\\g(y,z,t)=0,\\h(z,t)=0.\end{cases}$$

在什么条件下u是x,y的函数？求$\dfrac{\partial u}{\partial x}$，$\dfrac{\partial u}{\partial y}$.

解 考虑$\begin{cases}g\left(y,z,t\right)=0\\h\left(z,t\right)=0\end{cases}$，若$g\left(y,z,t\right)$，$h\left(z,t\right)$满足：

(1)在某一点$p_0=\left(y_0,z_0,t_0\right)$附近对各变量有一阶连续偏导数；

(2) $g\left(p_0\right)=h\left(p_0\right)=0$；

(3) $J=\left.\dfrac{\partial\left(g,h\right)}{\partial\left(z,t\right)}\right|_{p_0}\neq0$.

则在y_0点附近以上方程组唯一地确定一组函数$z=z\left(y\right)$，$t=t\left(y\right)$，且这组函数在y_0点附近连续可微，从而$u=f(x,y,u,v)=f\left(x,y,z\left(y\right),t\left(y\right)\right)$就是关于$x,y$的函数. 并有

$$\frac{\partial u}{\partial x}=\frac{\partial f}{\partial x},$$

$$\frac{\partial u}{\partial y}=\frac{\partial f}{\partial y}+\frac{\partial f}{\partial z}\frac{dz}{dy}+\frac{\partial f}{\partial t}\frac{dt}{dy}=\frac{\partial f}{\partial y}-\frac{1}{J}\frac{\partial f}{\partial z}\frac{\partial(g,h)}{\partial(y,t)}-\frac{\partial f}{\partial t}\frac{1}{J}\frac{\partial(g,h)}{\partial(z,y)},\ \text{其中}\ J=\frac{\partial(g,h)}{\partial(z,t)}.$$

7. 设函数$u=u(x)$由方程组

$$\begin{cases}u=f(x,y,z),\\ g(x,y,z)=0,\\ h(x,y,z)=0\end{cases}$$

所确定，求$\frac{du}{dx}$，$\frac{d^2u}{dx^2}$.

解 由于原方程组能确定函数$u=u(x)$，根据方程组中u的表达式可知$g(x,y,z)=0$和$h(x,y,z)=0$能确定y,z是x的函数，从而

$$\frac{dy}{dx}=-\frac{\partial(g,h)}{\partial(x,z)}\Big/\frac{\partial(g,h)}{\partial(y,z)},\quad \frac{dz}{dx}=-\frac{\partial(g,h)}{\partial(y,x)}\Big/\frac{\partial(g,h)}{\partial(y,z)}\qquad (*)$$

因此

$$\frac{du}{dx}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\cdot\frac{dy}{dx}+\frac{\partial f}{\partial z}\cdot\frac{dz}{dx}=\frac{\partial f}{\partial x}-\frac{\partial f}{\partial y}\cdot\frac{\partial(g,h)}{\partial(x,z)}\Big/\frac{\partial(g,h)}{\partial(y,z)}-\frac{\partial f}{\partial z}\cdot\frac{\partial(g,h)}{\partial(y,x)}\Big/\frac{\partial(g,h)}{\partial(y,z)}$$

再对$\frac{du}{dx}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\cdot\frac{dy}{dx}+\frac{\partial f}{\partial z}\cdot\frac{dz}{dx}$左右两边关于$x$求导，有

$$\begin{aligned}\frac{d^2u}{dx^2}&=\frac{d}{dx}\left(\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\cdot\frac{dy}{dx}+\frac{\partial f}{\partial z}\cdot\frac{dz}{dx}\right)\\&=\frac{\partial^2 f}{\partial x^2}+\frac{\partial^2 f}{\partial x\partial y}\cdot\frac{dy}{dx}+\frac{\partial^2 f}{\partial x\partial z}\cdot\frac{dz}{dx}+\left(\frac{\partial^2 f}{\partial y\partial x}+\frac{\partial^2 f}{\partial y^2}\cdot\frac{dy}{dx}+\frac{\partial^2 f}{\partial y\partial z}\cdot\frac{dz}{dx}\right)\frac{dy}{dx}+\frac{\partial f}{\partial y}\cdot\frac{d^2y}{dx^2}\\&\quad+\left(\frac{\partial^2 f}{\partial z\partial x}+\frac{\partial^2 f}{\partial z\partial y}\cdot\frac{dy}{dx}+\frac{\partial^2 f}{\partial z^2}\cdot\frac{dz}{dx}\right)\frac{dz}{dx}+\frac{\partial f}{\partial z}\cdot\frac{d^2z}{dx^2}\end{aligned}$$

其中$\frac{dy}{dx},\frac{dz}{dx}$由（*）式给出，而且根据（*）式知$\frac{d^2u}{dx^2}$的表达式中

$$\frac{d^2y}{dx^2}=\left[-\frac{\partial}{\partial x}\left(\frac{\partial(g,h)}{\partial(x,z)}\right)\cdot\frac{\partial(g,h)}{\partial(y,z)}+\frac{\partial(g,h)}{\partial(x,z)}\cdot\frac{\partial}{\partial x}\left(\frac{\partial(g,h)}{\partial(y,z)}\right)\right]\Big/\left(\frac{\partial(g,h)}{\partial(y,z)}\right)^2,$$

$$\frac{d^2z}{dx^2}=\left[-\frac{\partial}{\partial x}\left(\frac{\partial(g,h)}{\partial(y,x)}\right)\cdot\frac{\partial(g,h)}{\partial(y,z)}+\frac{\partial(g,h)}{\partial(y,x)}\cdot\frac{\partial}{\partial x}\left(\frac{\partial(g,h)}{\partial(y,z)}\right)\right]\Bigg/\left(\frac{\partial(g,h)}{\partial(y,z)}\right)^2.$$

8. 设 $z=z(x,y)$ 满足方程组

$$\begin{cases}f(x,y,z,t)=0,\\ g(x,y,z,t)=0.\end{cases}$$

求 dz.

解 由已知条件知方程组能确定函数组 $z=z(x,y),t=t(x,y)$，故

$$\frac{\partial z}{\partial x}=-\frac{\partial(f,g)}{\partial(x,t)}\Bigg/\frac{\partial(f,g)}{\partial(z,t)},\quad \frac{\partial z}{\partial y}=-\frac{\partial(f,g)}{\partial(y,t)}\Bigg/\frac{\partial(f,g)}{\partial(z,t)}$$

因而 $dz=\frac{\partial z}{\partial x}dx+\frac{\partial z}{\partial y}dy=\left[-\frac{\partial(f,g)}{\partial(x,t)}\Big/\frac{\partial(f,g)}{\partial(z,t)}\right]dx+\left[-\frac{\partial(f,g)}{\partial(y,t)}\Big/\frac{\partial(f,g)}{\partial(z,t)}\right]dy$.

9. 设

$$\begin{cases}u=f(x-ut,y-ut,z-ut),\\ g(x,y,z)=0.\end{cases}$$

求 $\frac{\partial u}{\partial x},\frac{\partial u}{\partial y}$. 这时 t 是自变量还是因变量？

解 在 $g(x,y,z)=0$ 两边对 x,y 求导，有 $g_1+g_3\frac{\partial z}{\partial x}=0$，$g_2+g_3\frac{\partial z}{\partial y}=0$，从而得

$$\frac{\partial z}{\partial x}=-\frac{g_1}{g_3},\frac{\partial z}{\partial y}=-\frac{g_2}{g_3}$$

所以

$$\begin{aligned}\frac{\partial u}{\partial x}&=f_1\left(1-t\frac{\partial u}{\partial x}\right)+f_2\left(-t\frac{\partial u}{\partial x}\right)+f_3\left(\frac{\partial z}{\partial x}-t\frac{\partial u}{\partial x}\right)\\&=f_1-f_1t\frac{\partial u}{\partial x}-f_2t\frac{\partial u}{\partial x}-f_3t\frac{\partial u}{\partial x}+f_3\left(-\frac{g_1}{g_3}\right)\end{aligned}$$

从中解出 $\frac{\partial u}{\partial x}=\frac{f_1g_3-f_3g_1}{g_3\left[1+t\left(f_1+f_2+f_3\right)\right]}$，同样由对称性得 $\frac{\partial u}{\partial y}=\frac{f_2g_3-f_3g_2}{g_3\left[1+t\left(f_1+f_2+f_3\right)\right]}$，

其中 t 是自变量.

10. 设 (x_0,y_0,z_0,u_0) 满足方程组

$$\begin{cases} f(x)+f(y)+f(z)=F(u), \\ g(x)+g(y)+g(z)=G(u), \\ h(x)+h(y)+h(z)=H(u), \end{cases}$$

这里假定所有的函数有连续的导数.

(1) 说出一个能在该点领域内确定 x,y,z 作为 u 的函数的充分条件；

(2) 在 $f(x)=x, g(x)=x^2, h(x)=x^3$ 的情形下，上述条件相当于什么？

解（1）设 $P_0=(x_0,y_0,z_0,u_0)$，则根据已知条件可知，当条件

(i) $\begin{cases} f(x_0)+f(y_0)+f(z_0)=F(u_0), \\ g(x_0)+g(y_0)+g(z_0)=G(u_0), \\ h(x_0)+h(y_0)+h(z_0)=H(u_0); \end{cases}$

(ii) $J|_{p_0}=\begin{vmatrix} f'(x) & f'(y) & f'(z) \\ g'(x) & g'(y) & g'(z) \\ h'(x) & h'(y) & h'(z) \end{vmatrix}\Bigg|_{p_0}=\begin{vmatrix} f'(x_0) & f'(y_0) & f'(z_0) \\ g'(x_0) & g'(y_0) & g'(z_0) \\ h'(x_0) & h'(y_0) & h'(z_0) \end{vmatrix}\neq 0.$

同时成立时，方程组就能在 P_0 的邻域内确定 x,y,z 作为 u 的函数.

（2）在 $f(x)=x, g(x)=x^2, h(x)=x^3$ 的情形下，上述条件相当于

(i) $x_0+y_0+z_0=F(u_0), {x_0}^2+{y_0}^2+{z_0}^2=G(u_0), {x_0}^3+{y_0}^3+{z_0}^3=H(u_0).$

(ii) $J=\begin{vmatrix} 1 & 1 & 1 \\ 2x_0 & 2y_0 & 2z_0 \\ 3{x_0}^2 & 3{y_0}^2 & 3{z_0}^2 \end{vmatrix}=6\begin{vmatrix} 1 & 1 & 1 \\ x_0 & y_0 & z_0 \\ {x_0}^2 & {y_0}^2 & {z_0}^2 \end{vmatrix}=6(y_0-x_0)(z_0-x_0)(z_0-y_0)\neq 0.$

即 x_0,y_0,z_0 两两不等.

11. 设 $x=u, y=\dfrac{u}{1+uv}, z=\dfrac{u}{1+uw}$，取 u,v 为新的自变量，w 为新的因变量，变换方程

$$x^2\frac{\partial z}{\partial x}+y^2\frac{\partial z}{\partial y}=z^2.$$

解 由 $x=u, y=\dfrac{u}{1+uv}$ 可得

$$\begin{cases} u=u(x,y)=x, \\ v=v(x,y)=\dfrac{1}{y}-\dfrac{1}{x}. \end{cases}$$

由于取 u,v 为新的自变量，w 为新的因变量，因而

$$z=\frac{u}{1+uw}=z(u,w)=z\left(u,w(u,v)\right)=z\left(u(x,y),w(u(x,y),v(x,y))\right)$$

因此

$$\begin{aligned}\frac{\partial z}{\partial x}&=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial z}{\partial w}\cdot\frac{\partial w}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial z}{\partial w}\cdot\frac{\partial w}{\partial v}\cdot\frac{\partial v}{\partial x}\\&=\frac{\partial}{\partial u}\left(\frac{u}{1+uw}\right)\cdot 1+\frac{\partial}{\partial w}\left(\frac{u}{1+uw}\right)\cdot\frac{\partial w}{\partial u}\cdot 1+\frac{\partial}{\partial w}\left(\frac{u}{1+uw}\right)\cdot\frac{\partial w}{\partial v}\cdot\left(\frac{1}{x^2}\right)\\&=\frac{1}{\left(1+uw\right)^2}\left(1-u^2\frac{\partial w}{\partial u}-\frac{\partial w}{\partial v}\right)\end{aligned}$$

同理

$$\frac{\partial z}{\partial y}=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial y}+\frac{\partial z}{\partial w}\cdot\frac{\partial w}{\partial u}\cdot\frac{\partial u}{\partial y}+\frac{\partial z}{\partial w}\cdot\frac{\partial w}{\partial v}\cdot\frac{\partial v}{\partial y}=0+0+\frac{\partial}{\partial w}\left(\frac{u}{1+uw}\right)\cdot\frac{\partial w}{\partial v}\cdot\left(-\frac{1}{y^2}\right)$$

$$=\frac{u^2}{\left(1+uw\right)^2}\cdot\frac{\partial w}{\partial v}\cdot\frac{1}{y^2}$$

代入方程 $x^2\frac{\partial z}{\partial x}+y^2\frac{\partial z}{\partial y}=z^2$ 得

$$\frac{u^2}{\left(1+uw\right)^2}\left(1-u^2\frac{\partial w}{\partial u}-\frac{\partial w}{\partial v}\right)+\frac{u^2}{\left(1+uw\right)^2}\cdot\frac{\partial w}{\partial v}=\left(\frac{u}{1+uw}\right)^2$$

化简后得到 $u^2\frac{\partial w}{\partial u}=0$，这就是方程 $x^2\frac{\partial z}{\partial x}+y^2\frac{\partial z}{\partial y}=z^2$ 用 $w=w(u,v)$ 表示的新形式.